

```
$ python3 day01.py
```

Python Day01

0000000000

```
>>> print("00000000000000")
```

```
00000000000000
```



```
$ cat agenda.md
```



1. Python Day01 — □□□□□□□□□□

2. □□□□

3. □□□□□□□□□□

4. □□□□□□□□

5. □□□□□□□□□□□□

6. □□

7. □□□□

8. □□□□□□

9. □□□□□□□□

10. Output

11. □□□□□□□□□□□□

12. □□□□□□□

13. f-string □□□□□□



Python

(type)

10

int
□□

12.3

float
□□□

'hi'

str
□□

True

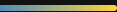
bool
□□

None

NoneType
□□

```
>>> type(10) # <class 'int'>
>>> type('hi') # <class 'str'>
```





```
# [] [] [] [] []  
'123' ~ [] []  
123 ~ [] []  
'123' == 123 → False
```

'123'



123



```
>>> '123' == 123  
False # [] [] [] [] [] [] [] []
```

 [] [] [] [] []     input() [] [] [] []




[Python](#)
[C++](#)
[C](#)
[Java](#)
[JavaScript](#)
[PHP](#)
[Go](#)
[Ruby](#)
[Swift](#)
[Kotlin](#)
[TypeScript](#)
[R](#)
[Perl](#)
[Lua](#)
[Python 3](#)
[Python 2](#)
[Python 1](#)

🤔

1. $\frac{1}{2}$ 2. $\frac{1}{3}$ 3. $\frac{1}{4}$ 4. $\frac{1}{5}$ 5. $\frac{1}{6}$ 6. $\frac{1}{7}$ 7. $\frac{1}{8}$ 8. $\frac{1}{9}$ 9. $\frac{1}{10}$ 10. $\frac{1}{11}$ 11. $\frac{1}{12}$ 12. $\frac{1}{13}$ 13. $\frac{1}{14}$ 14. $\frac{1}{15}$ 15. $\frac{1}{16}$ 16. $\frac{1}{17}$ 17. $\frac{1}{18}$ 18. $\frac{1}{19}$ 19. $\frac{1}{20}$ 20. $\frac{1}{21}$ 21. $\frac{1}{22}$ 22. $\frac{1}{23}$ 23. $\frac{1}{24}$ 24. $\frac{1}{25}$ 25. $\frac{1}{26}$ 26. $\frac{1}{27}$ 27. $\frac{1}{28}$ 28. $\frac{1}{29}$ 29. $\frac{1}{30}$ 30. $\frac{1}{31}$ 31. $\frac{1}{32}$ 32. $\frac{1}{33}$ 33. $\frac{1}{34}$ 34. $\frac{1}{35}$ 35. $\frac{1}{36}$ 36. $\frac{1}{37}$ 37. $\frac{1}{38}$ 38. $\frac{1}{39}$ 39. $\frac{1}{40}$ 40. $\frac{1}{41}$ 41. $\frac{1}{42}$ 42. $\frac{1}{43}$ 43. $\frac{1}{44}$ 44. $\frac{1}{45}$ 45. $\frac{1}{46}$ 46. $\frac{1}{47}$ 47. $\frac{1}{48}$ 48. $\frac{1}{49}$ 49. $\frac{1}{50}$ 50. $\frac{1}{51}$ 51. $\frac{1}{52}$ 52. $\frac{1}{53}$ 53. $\frac{1}{54}$ 54. $\frac{1}{55}$ 55. $\frac{1}{56}$ 56. $\frac{1}{57}$ 57. $\frac{1}{58}$ 58. $\frac{1}{59}$ 59. $\frac{1}{60}$ 60. $\frac{1}{61}$ 61. $\frac{1}{62}$ 62. $\frac{1}{63}$ 63. $\frac{1}{64}$ 64. $\frac{1}{65}$ 65. $\frac{1}{66}$ 66. $\frac{1}{67}$ 67. $\frac{1}{68}$ 68. $\frac{1}{69}$ 69. $\frac{1}{70}$ 70. $\frac{1}{71}$ 71. $\frac{1}{72}$ 72. $\frac{1}{73}$ 73. $\frac{1}{74}$ 74. $\frac{1}{75}$ 75. $\frac{1}{76}$ 76. $\frac{1}{77}$ 77. $\frac{1}{78}$ 78. $\frac{1}{79}$ 79. $\frac{1}{80}$ 80. $\frac{1}{81}$ 81. $\frac{1}{82}$ 82. $\frac{1}{83}$ 83. $\frac{1}{84}$ 84. $\frac{1}{85}$ 85. $\frac{1}{86}$ 86. $\frac{1}{87}$ 87. $\frac{1}{88}$ 88. $\frac{1}{89}$ 89. $\frac{1}{90}$ 90. $\frac{1}{91}$ 91. $\frac{1}{92}$ 92. $\frac{1}{93}$ 93. $\frac{1}{94}$ 94. $\frac{1}{95}$ 95. $\frac{1}{96}$ 96. $\frac{1}{97}$ 97. $\frac{1}{98}$ 98. $\frac{1}{99}$ 99. $\frac{1}{100}$ 100. $\frac{1}{101}$ 101. $\frac{1}{102}$ 102. $\frac{1}{103}$ 103. $\frac{1}{104}$ 104. $\frac{1}{105}$ 105. $\frac{1}{106}$ 106. $\frac{1}{107}$ 107. $\frac{1}{108}$ 108. $\frac{1}{109}$ 109. $\frac{1}{110}$ 110. $\frac{1}{111}$ 111. $\frac{1}{112}$ 112. $\frac{1}{113}$ 113. $\frac{1}{114}$ 114. $\frac{1}{115}$ 115. $\frac{1}{116}$ 116. $\frac{1}{117}$ 117. $\frac{1}{118}$ 118. $\frac{1}{119}$ 119. $\frac{1}{120}$ 120. $\frac{1}{121}$ 121. $\frac{1}{122}$ 122. $\frac{1}{123}$ 123. $\frac{1}{124}$ 124. $\frac{1}{125}$ 125. $\frac{1}{126}$ 126. $\frac{1}{127}$ 127. $\frac{1}{128}$ 128. $\frac{1}{129}$ 129. $\frac{1}{130}$ 130. $\frac{1}{131}$ 131. $\frac{1}{132}$ 132. $\frac{1}{133}$ 133. $\frac{1}{134}$ 134. $\frac{1}{135}$ 135. $\frac{1}{136}$ 136. $\frac{1}{137}$ 137. $\frac{1}{138}$ 138. $\frac{1}{139}$ 139. $\frac{1}{140}$ 140. $\frac{1}{141}$ 141. $\frac{1}{142}$ 142. $\frac{1}{143}$ 143. $\frac{1}{144}$ 144. $\frac{1}{145}$ 145. $\frac{1}{146}$ 146. $\frac{1}{147}$ 147. $\frac{1}{148}$ 148. $\frac{1}{149}$ 149. $\frac{1}{150}$ 150. $\frac{1}{151}$ 151. $\frac{1}{152}$ 152. $\frac{1}{153}$ 153. $\frac{1}{154}$ 154. $\frac{1}{155}$ 155. $\frac{1}{156}$ 156. $\frac{1}{157}$ 157. $\frac{1}{158}$ 158. $\frac{1}{159}$ 159. $\frac{1}{160}$ 160. $\frac{1}{161}$ 161. $\frac{1}{162}$ 162. $\frac{1}{163}$ 163. $\frac{1}{164}$ 164. $\frac{1}{165}$ 165. $\frac{1}{166}$ 166. $\frac{1}{167}$ 167. $\frac{1}{168}$ 168. $\frac{1}{169}$ 169. $\frac{1}{170}$ 170. $\frac{1}{171}$ 171. $\frac{1}{172}$ 172. $\frac{1}{173}$ 173. $\frac{1}{174}$ 174. $\frac{1}{175}$ 175. $\frac{1}{176}$ 176. $\frac{1}{177}$ 177. $\frac{1}{178}$ 178. $\frac{1}{179}$ 179. $\frac{1}{180}$ 180. $\frac{1}{181}$ 181. $\frac{1}{182}$ 182. $\frac{1}{183}$ 183. $\frac{1}{184}$ 184. $\frac{1}{185}$ 185. $\frac{1}{186}$ 186. $\frac{1}{187}$ 187. $\frac{1}{188}$ 188. $\frac{1}{189}$ 189. $\frac{1}{190}$ 190. $\frac{1}{191}$ 191. $\frac{1}{192}$ 192. $\frac{1}{193}$ 193. $\frac{1}{194}$ 194. $\frac{1}{195}$ 195. $\frac{1}{196}$ 196. $\frac{1}{197}$ 197. $\frac{1}{198}$ 198. $\frac{1}{199}$ 199. $\frac{1}{200}$ 200. $\frac{1}{201}$ 201. $\frac{1}{202}$ 202. $\frac{1}{203}$ 203. $\frac{1}{204}$ 204. $\frac{1}{205}$ 205. $\frac{1}{206}$ 206. $\frac{1}{207}$ 207. $\frac{1}{208}$ 208. $\frac{1}{209}$ 209. $\frac{1}{210}$ 210. $\frac{1}{211}$ 211. $\frac{1}{212}$ 212. $\frac{1}{213}$ 213. $\frac{1}{214}$ 214. $\frac{1}{215}$ 215. $\frac{1}{216}$ 216. $\frac{1}{217}$ 217. $\frac{1}{218}$ 218. $\frac{1}{219}$ 219. $\frac{1}{220}$ 220. $\frac{1}{221}$ 221. $\frac{1}{222}$ 222. $\frac{1}{223}$ 223. $\frac{1}{224}$ 224. $\frac{1}{225}$ 225. $\frac{1}{226}$ 226. $\frac{1}{227}$ 227. $\frac{1}{228}$ 228. $\frac{1}{229}$ 229. $\frac{1}{230}$ 230. $\frac{1}{231}$ 231. $\frac{1}{232}$ 232. $\frac{1}{233}$ 233. $\frac{1}{234}$ 234. $\frac{1}{235}$ 235. $\frac{1}{236}$ 236. $\frac{1}{237}$ 237. $\frac{1}{238}$ 238. $\frac$

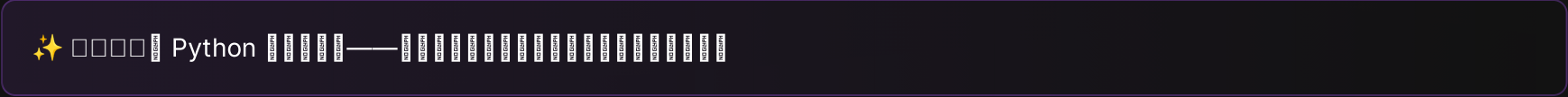


□ □ □ □ □ □ □ □ □ □ □ □



□ □ □ □ □ □ □ □ □ □

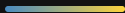
if, for, while



PART 03

Output

```
❏ print() [] [] [] [] []
```



```
1 print('hi')
2 print('hi')      # 00000000 hi 000
```

00000

```
1 hi
2 hi
```



```
1 # 😞 000 - 000
2 name = 'Lucas'
3 age = 18
4 print('Hello,', name, 'you are', age, 'years old')
```

```
1 # 😊 0 f-string - 0000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! You are {age} years old.')
```

```
1 # 🌟 f-string 0000000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! Next year you will be {age + 1}')
```



print()



□□□□ □□□□ □□□□ □□□□

```
1 print('hi')
2 print('hi')
3 # :
4 # hi
5 # hi
```

🔑 end — '\n' · sep — ' ' □

f-string

f-string

```
1  # 0000
2  print(f'{{}}')
3
4  # 0000000000000000
5  pi = 3.14159265
6  print(f'π ≈ {pi:.2f}')    # π ≈ 3.14
7  print(f'π ≈ {pi:.4f}')    # π ≈ 3.1416
8
9  # 0000
10 print(f'{{42:05d}}')      # 00042
```

0000

```
{{:0}}
:.2f  - 0000
:05d  - 0000
```

0000

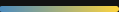
0000000000000000

```
1  pi = 3.14159265
2  print(f'π ≈ {pi:.2f}')
3  print(f'π ≈ {pi:.4f}')
4  print(f'{{42:05d}}')
```

PART 04

Input

□□□□□□□□□□



```

sequenceDiagram
    participant P as Person
    participant L as Laptop
    participant M as Monitor
    P->>L: input()
    L->>L: 
    L->>M: print()
    M-->>P: 
  
```




`input()` □□□□□□□□□□□□

```
1 n = input()
2 nn = input('□□□□□□□□')
```

⚠ `input()` □□□□□□□□□□□□□□□□
[REDACTED] '123' != 123 [REDACTED]



```
1 # ✎ □□□□□□□□□□
2 name = input('□□□□□ ')
3 print(f'Hello, {name}!')
4
5 # input() □□□□□□□□
6 age_str = input('□□□ ')
7 print(type(age_str)) # <class 'str'>
8
9 age = int(age_str) # □□□□
10 print(f'□□□□ {age + 1} □□')
```



□□□□□□□□□□□□□□□□

sys.stdin while + try □□□□

```
1  import sys
2
3  for line in sys.stdin:
4      line = line.strip()
5      print(line)
```

 sys.stdin

□□□
□□□□□□

 while+try

□□□□
□□□□□□

 □□□□

□□□
□□□□□□



a.py ☐

--	--	--	--	--	--

□ □ □ □ □

in.txt ☐

```
1 Everything will get better 12345
```



□□□□□□□□

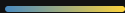
  in.txt □□□□□□□□□□□□□□□□

```
1 import sys
2
3 # UTF-8
4 sys.stdin.reconfigure(encoding='utf-8')
5
6 for line in sys.stdin:
7     line = line.strip()
8     print(line)
```

□□□□□□□□

```
1 python a.py < in.txt > out.txt
```

 UTF-8 Windows□□□□□□



```
1 n = 10
```

```
n + 2 → 12
```

```
n - 2 → 8
```

```
n * 2 → 20
```

```
n / 2 → 5.0 ← float
```

```
n // 2 → 5
```

```
n ** 6 → 1000000
```

```
n % 6 → 4 ← 10 ÷ 6 残り
```

💡 /  float `//` `□□□□□□□□□□`

□ □ □ □ □ □ □

□ □ □ □ □ □ □ □ □ □

□ □ □ □ □ □ □ □ □ □

```
1  x = 10
2  x = x + 5    # x == 15
3  x = x * 2    # x == 30
4  print(x)
```

⚡ `x += 1` `10` `11` `12` `13` `14` `15` `16` `17` `18` `19` `20` `21` `22` `23` `24` `25` `26` `27` `28` `29` `30` `31` `32` `33` `34` `35` `36` `37` `38` `39` `40` `41` `42` `43` `44` `45` `46` `47` `48` `49` `50` `51` `52` `53` `54` `55` `56` `57` `58` `59` `60` `61` `62` `63` `64` `65` `66` `67` `68` `69` `70` `71` `72` `73` `74` `75` `76` `77` `78` `79` `80` `81` `82` `83` `84` `85` `86` `87` `88` `89` `90` `91` `92` `93` `94` `95` `96` `97` `98` `99` `100`

□□□□□□□□

□□□□□□□□□□

■ ✓ True □□□

■ ✗ False □□□

□□□	□□
<	□□
<=	□□□□
>	□□
>=	□□□□
==	□□
!=	□□□

□ □

```
1 10 <= 60      # True
2 123 == 123     # True
3 20 == 40       # False
4 '123' == 123   # False ⚠
```

⚠ '123' 123
False

Logical Operators

Python Logical Operators

Operator	Description
<code>and</code>	Both operands must be true
<code>or</code>	At least one operand must be true
<code>not</code>	Reverse the logical state of the operand

```
1 age = 18
2 has_id = True
3
4 age >= 18 and has_id    # True
5 age < 18 or has_id     # True
6 not has_id             # False
```

🧠 `and` = Both operands must be true · `or` = At least one operand must be true · `not` = Reverse the logical state of the operand

Truthy / Falsy □□□□□□

Python True / False □□□□□□

```
1 bool(0)           # False
2 bool(1)           # True
3 bool("")          # False
4 bool("hi")        # True
5 bool(None)        # False
6 bool([])          # False
```

Falsy False

- 0
- " "
- None
- []
- {}

Truthy True

- 0
-
-

if

```
1 age = 18
2
3 if age >= 18:
4     print("0000")
```

 Python 000000000000000000

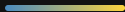
if / elif / else

□□□□□□

```
1  score = 85
2
3  if score >= 90:
4      print("A")
5  elif score >= 80:
6      print("B")
7  elif score >= 70:
8      print("C")
9  else:
10     print("F")
```



if score >= 90: print("A")
elif score >= 80: print("B")
elif score >= 70: print("C")
else: print("F")



```
1 username = input("000")
2 password = input("000")
3
4 if username == "admin" and password == "1234":
5     print("0000")
6 else:
7     print("0000")
```

🔑 000000input + 00 + and 0000



- Python □□□□□int / float / str / bool / None

- `input()` □□□□□□□□ □□

- `print()` □□□□□□□□ f-string □□□

- `/` □ `//` □ `%` □□□□□□□□

- □□□□□□□□ + □□□□□□□□

- `if` / `elif` / `else` □□□□

